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Bridging Process and Practice: A Blueprint for Integrating the EMAD and BMAD Frameworks

Deconstructing the EMAD Ecosystem: A Developer-Centric Enhancement Framework

The EMAD framework, officially described as an "Ecosystem for BMad Method AI Development," represents a developer-centric approach to augmenting software development processes through artificial intelligence [1]. Hosted on GitHub under the repository `huggingfacer04/EMAD`, its primary design philosophy appears to be the enhancement of a developer's existing workflow rather than a complete overhaul of the development lifecycle [1]. The framework's most prominent feature is its comprehensive Visual Studio Code (VS Code) extension, which serves as the central interface for its functionalities [1]. This integration positions EMAD not as a standalone platform but as a powerful toolkit designed to operate within the daily environment of a software engineer, aiming to boost productivity and streamline interactions with AI models. While the provided sources lack exhaustive documentation on every module, the available context allows for a detailed analysis of its structure, purpose, and potential role within the AI-driven development ecosystem.

The core mechanism of EMAD is its VS Code extension, which embeds a suite of AI-powered tools directly into the editor [1]. This strategy prioritizes developer experience by minimizing context switching and providing assistance where the work is being done. The extension integrates AI agents designed to offer holistic support throughout the software development cycle [1]. The description "Ecosystem for BMad Method AI Development" suggests a deep, intentional relationship with the Breakthrough Method for Agile AI-Driven Development (BMAD) [1]. This implies that EMAD may provide a more integrated, accessible, or enhanced implementation of the BMAD methodology specifically tailored for the VS Code environment. It could potentially act as a bridge, translating the high-level, YAML-driven workflows of BMAD into concrete, interactive tasks within the IDE. For instance, it might automate the execution of BMAD commands, visualize workflow states, or provide contextual prompts and code snippets based on the current stage of a BMAD project. This symbiotic relationship is crucial; EMAD enhances

the developer's ability to execute the BMAD process effectively, while BMAD provides the overarching structure and logic that EMAD helps to operationalize.

Regarding its technical architecture, the precise internal components of the EMAD framework are not fully detailed in the provided materials. However, its function as a VS Code extension indicates it is built using web technologies such as TypeScript, HTML, and CSS, leveraging the VS Code Extension API for its integrations [136]. Its interaction with external AI models would likely involve standard REST APIs or WebSocket connections to language model providers. The mention of a "comprehensive VS Code extension with integrated AI agents" points to a modular design where different functionalities are packaged as distinct tools or panels within the editor [1]. This aligns with the broader trend of building specialized, opinionated tools that integrate deeply with popular IDEs to provide a seamless user experience. The focus on being an "ecosystem" also hints at extensibility, suggesting that third-party developers or users might be able to contribute their own AI agents or tools to the framework, expanding its capabilities over time.

Installation and deployment of EMAD are straightforward, mirroring the standard procedure for VS Code extensions. Users would typically browse the VS Code Extensions Marketplace, search for "EMAD," and install it with a single click. Alternatively, if distributed via its GitHub repository, users could download the source code and manually install it. This low-friction installation process is a significant advantage, drastically lowering the barrier to entry compared to frameworks that require complex local environment setups or server deployments. The entire toolset becomes available immediately upon installation, ready to be configured for a specific project or organization's needs. This contrasts sharply with MLOps platforms like Kubeflow or MLflow, which are designed for managing entire machine learning pipelines and often require significant infrastructure provisioning and configuration [116][118]. EMAD's deployment model is inherently lightweight and targeted at individual developers or small teams working within the VS Code environment.

In terms of workflow management, EMAD's approach is likely to be less explicitly defined than that of BMAD. While BMAD relies on a formal, stateful system of YAML files (`workflow.yaml`) to define variables, dependencies, and track progress across a multi-agent team [60], EMAD probably focuses on more immediate, task-oriented automation. For example, it might manage workflows for generating code snippets, writing unit tests, or performing documentation updates. These workflows would be triggered by user actions within the editor (e.g., a command, a button click, or a keyboard shortcut) rather than being part of a long-running, centralized project plan. The challenge of managing state in agentic workflows—tracking progress, handling errors, and ensuring reliability—

is a known issue in the field [62] [66]. EMAD's solution would likely be more implicit, managing the state of the current editing session or the state of a single, short-lived task, rather than the complex, multi-step state of an entire project managed by multiple AI agents [59].

The intended audience for EMAD is clearly the software developer who works within the VS Code ecosystem. This includes individual contributors, small startup teams, and potentially larger organizations looking to empower their engineering workforce with AI-assisted tools without mandating a radical change in their established workflows. By focusing on the developer's immediate context, EMAD aims to reduce cognitive load and increase efficiency on a per-task basis. This is different from frameworks targeting data scientists, like those in the MLOps space (e.g., Kubeflow, MLflow), which manage experiments, track metrics, and deploy models at scale [117] [122]. EMAD is not about managing the ML lifecycle but about accelerating the coding and debugging phases of traditional software development. Its value proposition lies in making the developer's day-to-day work faster and more effective.

Regarding adoption trends, the provided information does not contain quantitative metrics such as GitHub stars, forks, or user statistics for the EMAD framework [93]. The existence of the `huggingfacer04/EMAD` repository is noted, indicating some level of public presence, but its popularity, maturity, and community engagement remain unknown from the given sources [1]. In contrast, the BMAD methodology shows signs of a dedicated community, with active discussion forums, masterclasses, and ongoing development reflected in GitHub issues and documentation [20] [21] [24] [6]. The adoption of any AI framework depends heavily on factors like developer experience, perceived utility, and community support [94] [126]. Without data on EMAD's usage, it is difficult to assess its market penetration. However, its developer-centric model gives it a clear niche. If the VS Code extension proves to be intuitive, reliable, and genuinely productive, it could attract a loyal following among VS Code users seeking to leverage AI in their daily work. The success of similar tools often hinges on solving a specific pain point for a well-defined audience, a strategy that EMAD appears to be pursuing [98]. Further investigation would be required to understand its actual adoption rate, contribution activity, and user feedback to gauge its real-world impact and viability.

The BMAD Methodology: A Structured, Agentic Approach to Software Development

The BMAD Method, or Breakthrough Method for Agile AI-Driven Development, stands as a comprehensive, process-oriented framework designed to impose structure and predictability on the chaotic nature of AI-assisted software development [14] [163]. Unlike tool-focused ecosystems like EMAD, BMAD is fundamentally a methodology that defines a rigorous workflow centered around specialized AI agents acting as a virtual agile team [35] [102]. Its core philosophy is to simulate a human development team, assigning distinct roles such as Analyst, Product Manager (PM), Architect, Scrum Master (SM), Developer, and Quality Assurance (QA) to different AI agents [16] [34]. This "team-simulating" approach is its defining characteristic, aiming to combat the unreliability and inconsistency often associated with ad-hoc AI coding, colloquially known as "vibe coding" [24] [35]. The goal is to transform AI development from an unpredictable experiment into a deterministic, repeatable process that yields production-grade results [102] [269].

A cornerstone of the BMAD methodology is its front-loaded planning phase, which concentrates AI usage on creating comprehensive artifacts before any code is written [15]. This spec-oriented approach involves the creation of detailed documents like a Product Requirements Document (PRD) and a high-level architecture plan [245] [267]. The rationale is that by investing AI effort upfront in planning, the subsequent development stages become more focused, efficient, and less prone to rework [15]. This disciplined, top-down planning is a direct response to the "planning inconsistency" that plagues less structured methods, making the overall development process more reliable and predictable [35]. This emphasis on specifications is so central that BMAD has been described as a form of specification-driven development, reviving a concept reminiscent of the Waterfall model but adapted for the modern era of generative AI [270].

The heart of the BMAD framework is its multi-agent orchestration system. Each AI agent is not a generic prompt template but a highly specialized entity with a deep understanding of its role and the overall BMAD architecture [28] [102]. The workflow is meticulously designed to mimic a real-world agile sprint. For example, a product idea is first analyzed by an Analyst agent, leading to a PRD created by a PM agent. An Architect agent then designs the system, after which the Scrum Master (SM) agent breaks down the architecture into user stories [67]. These stories are then sequentially assigned to a Developer (DEV) agent for implementation and finally passed to a QA agent for testing and review [67]. This structured flow ensures that tasks are completed in a logical order and that quality gates are enforced at each stage. The orchestrator, referred to as the

BMad Builder, is a master builder agent responsible for overseeing all creation workflows and maintaining consistency with BMAD conventions [28]. This systematic, role-based collaboration is what distinguishes BMAD from simpler, single-agent prompting strategies [163].

To manage the complexity of these multi-stage workflows, BMAD v6 introduced a formal state management system built around a set of YAML files [60]. The primary file, `workflow.yaml`, declares variables, file paths, and metadata for the project [60]. Other Markdown files, such as `bmad-workflow-bmm-code-review.md`, serve as containers for specific workflow steps, documenting the state and progress of tasks like code reviews [72] [74]. This explicit, file-based state management provides transparency and traceability, allowing developers to easily pinpoint where a workflow failed or to resume a task from a specific step [66]. However, this rigor comes at a cost. The complexity of managing these files has been identified as a significant barrier to adoption, representing a direct trade-off between the quality and reliability of the output and the ease of use of the framework [69]. The community has acknowledged that the workflow's intricacy is a necessary compromise for achieving higher quality and predictability in AI-generated code [69]. Furthermore, the manual execution of sequential commands required by early versions of BMAD was found to be error-prone and inefficient, highlighting the need for better automation and integration with developer tools [58].

The BMAD framework is designed to be extensible and adaptable to various domains. It supports a range of "expansion packs" that provide specialized agents, workflows, and templates for specific areas of software development [26]. Examples include packs for cloud infrastructure, data architecture, API development, and even AI/ML engineering [26] [29]. This modularity allows teams to tailor the framework to their specific technical stack and project requirements. Perhaps more significantly, BMAD introduces the BMAD Builder, a powerful module that empowers users to create their own custom modules, agents, and workflows [13] [244]. This capability moves BMAD beyond a rigid, predefined methodology towards a truly customizable framework, enabling organizations to codify their unique development practices into reusable, AI-executable components. This flexibility is crucial for long-term adoption, as it allows the framework to evolve alongside a team's changing needs and expertise.

Strategically, BMAD is positioned as a robust solution for entire agile teams, contrasting with lighter-weight frameworks like GitHub's Spec Kit, which are often seen as more suitable for individual developers [31]. BMAD's strength lies in its ability to facilitate collaboration and shared understanding among different roles within a team—PMs,

architects, and developers—by providing a common, structured language and process mediated by AI agents [31]. It is presented as the framework for reclaiming control in AI development, establishing a clear, auditable trail of decisions and artifacts from conception to completion [33]. By simulating a human team, BMAD provides a mental model that is familiar to software professionals, making the transition to AI-assisted development smoother and more intuitive [20]. The framework's ultimate goal is to make AI development systematic, reproducible, and scalable across an organization, moving beyond one-off experiments to become a core part of the engineering culture [65].

Comparative Analysis: EMAD, BMAD, and the Broader AI Agent Landscape

Choosing an appropriate framework for AI-driven development requires navigating a diverse and rapidly evolving landscape of tools, each with distinct philosophies, target audiences, and trade-offs. The BMAD methodology and the EMAD ecosystem represent two different paradigms within this space, and understanding their positioning relative to foundational libraries like LangChain and high-level collaboration frameworks like CrewAI is essential for strategic decision-making. A comparative analysis reveals a spectrum of abstraction, from low-level component building to opinionated, end-to-end process management.

At the foundation of many AI applications are libraries like LangChain and LlamaIndex. These are not complete frameworks but rather powerful, flexible toolkits for building Retrieval-Augmented Generation (RAG) systems and agentic applications [3] [288]. LangChain is a versatile, modular platform supporting a wide array of use cases, including chains and agents, while LlamaIndex is more specialized, excelling in streamlined search-and-retrieval and offering deeper coverage of advanced RAG techniques [4] [5]. They provide the "bricks" and "toolkit" for developers to construct their own bespoke AI systems [44]. Frameworks like CrewAI and Microsoft's AutoGen build upon this foundation, offering higher levels of abstraction focused specifically on multi-agent collaboration [39] [41]. CrewAI, for example, prioritizes speed and ease of use for setting up quick multi-agent collaborations, while AutoGen provides the fundamental components for building agent-to-agent conversational patterns [39] [127]. LangGraph, another key player, emphasizes control and robustness, making it suitable for complex branching workflows, a philosophy that differs from CrewAI's focus on simplicity [79].

The BMAD methodology occupies a position far above these foundational and collaborative frameworks in the hierarchy of abstraction. It is not merely a library for connecting models to tools; it is a complete, top-down development process [163]. Its focus is not just on building a multi-agent system but on structuring the entire software development lifecycle around such a system. It imposes a strict workflow, defined by planning documents and orchestrated by role-specific agents, aiming for determinism and quality at the expense of initial complexity [69] [102]. BMAD is comparable in ambition to MLOps platforms like Kubeflow or MLflow, but with a narrower focus on the application code generation aspect of the development process rather than the entire ML lifecycle [116] [118]. It seeks to answer the question, "How do we systematically and reliably produce high-quality software using AI?" rather than "How do we connect an LLM to a database?"

The EMAD framework carves out a unique niche that is complementary to BMAD. Instead of prescribing a full development process, EMAD acts as a tactical enhancement layer, delivered primarily as a VS Code extension [1]. Its purpose is to improve the developer's immediate workflow and productivity. While BMAD tells you *what* to do and *how* the AI team should collaborate, EMAD provides the *tools* to do it more efficiently within the developer's preferred environment. It fits into the same high-level category as CrewAI or LangGraph in that it deals with multi-agent systems, but its scope is intentionally limited to the developer's IDE [41]. The synergy is evident: BMAD provides the strategic blueprint and orchestration, while EMAD provides the ergonomic, hands-on tools to execute that blueprint. This makes them partners rather than competitors. EMAD could serve as a superior execution layer for BMAD, automating tedious command-line interactions and providing a more intuitive visual interface for managing the workflow [58] [75].

The table below summarizes the key differences between these frameworks and methodologies, highlighting their distinct characteristics and ideal use cases.

Feature	BMAD Method	EMAD Framework	LangChain / LlamaIndex	CrewAI / LangGraph
Core Philosophy	Process-driven, top-down, spec-oriented [15]	Tool-driven, bottom-up, DX-focused [1]	Foundational libraries for RAG/agentic systems [253]	High-level frameworks for multi-agent collaboration [127]
Primary User	Entire development team (PM, Dev, QA) [31]	Individual developer / VS Code user [1]	Application builders, researchers [39]	Developers building multi-agent systems [46]
Workflow Control	Highly structured, stateful YAML files [60]	Integrated within IDE, likely less explicit state management	Flexible, requires manual orchestration [5]	Flexible, with options for state machines (LangGraph) [79]
Agent Model	Role-specific, team-simulating agents [34]	Integrated enhancement agents [1]	General-purpose agents and chains [5]	Role/task-based crew coordination [127]
Adoption Barrier	Higher learning curve due to complexity [69]	Low (if well-integrated into VS Code) [1]	Moderate (requires understanding of concepts)	Low to moderate (prioritizes ease of use) [126]

This comparison underscores a fundamental trade-off in the AI development toolchain. On one end, frameworks like LangChain offer maximum flexibility and power for building novel, custom applications, but demand significant developer expertise and effort [83]. On the other end, methodologies like BMAD offer maximum structure and predictability, reducing risk and increasing reliability, but at the cost of rigidity and a steeper learning curve [69]. EMAD sits in a promising middle ground, aiming to lower the friction of adopting a structured methodology like BMAD by providing a polished, integrated developer experience. For the creator of 'Teams for new projects,' this analysis suggests that a hybrid approach—one that combines the strategic rigor of BMAD with the tactical, UX-focused execution of an EMAD-like tool—represents a compelling opportunity to address the market's need for both control and productivity in AI-assisted development.

Integration Strategy: Synthesizing Process Rigor with Tactical Execution

The strategic imperative for creating a new project by integrating EMAD and the BMAD methodology stems from the recognition that they are not competitors but complementary forces, each addressing a different facet of the AI-driven development challenge. BMAD provides the essential strategic backbone—a structured, process-oriented framework that enforces discipline, predictability, and quality through its agentic team simulation [35] [102]. EMAD offers the tactical execution layer—an integrated, developer-centric ecosystem designed to enhance productivity and streamline workflows directly within the developer's primary environment, VS Code [1]. A successful synthesis would create a unified system that leverages BMAD's "what" and "why" to guide the project, while EMAD's "how" provides the efficient, hands-on means of implementation. This hybrid model promises to mitigate the primary weaknesses of each component: the complexity and steep learning curve of BMAD [69] and the potential lack of overarching structure in more loosely defined tools.

The core of the integration strategy lies in using the BMAD methodology as the foundational engine and the EMAD framework as the user-facing interface and execution layer. The BMAD framework's stateful, YAML-driven workflow engine (`workflow.yaml`, `*.md`) would serve as the authoritative source of truth for the entire project's plan, status, and context [60]. This file would define the high-level objectives, the sequence of epics and stories, the assignment of tasks to agent roles, and the variables needed for the

AI agents' prompts. The EMAD VS Code extension would then act as a sophisticated client for this engine. Instead of requiring developers to manually navigate and execute BMAD commands from a terminal—a process noted for being error-prone and inefficient [58]—the EMAD extension would provide a graphical, interactive interface. Developers could view the current state of the workflow, trigger specific BMAD commands with a button click, see the outputs of AI agents in a dedicated panel, and receive contextual suggestions based on the project's current state.

This integrated approach directly addresses several known pain points in the current AI development landscape. First, it solves the discoverability problem highlighted by the BMAD community, where users struggle to remember workflow shortcuts like `<em>prd` or `</em>create-story` [75]. The EMAD extension would surface these actions through menus, buttons, and intelligent auto-completion within the editor, making the powerful features of BMAD much more accessible to new users. Second, it mitigates the complexity of managing BMAD's intricate YAML and Markdown files. The extension could provide a visual editor for the workflow, allowing developers to drag-and-drop tasks, link user stories, and configure agent parameters through forms rather than raw text, reducing the likelihood of syntax errors like the "multiline key may not be an implicit key" YAML parsing bug reported in the community [72]. Third, it creates a seamless feedback loop. When an AI agent generates code, the EMAD extension can automatically format it, run linters, and suggest running corresponding unit tests, all within the familiar VS Code environment. The results of these automated checks could then be fed back into the BMAD workflow, updating the state and potentially triggering the next step (e.g., passing a story to the QA agent).

Functionally, the integration would manifest in several key areas. For **planning**, the EMAD extension could have a dedicated panel for managing the BMAD specification documents (PRD, architecture). It could provide templates, collaborative editing features, and AI-powered summarization or analysis of these documents. For **development**, the extension would handle the interaction with the BMAD-developer agent. It could automatically generate boilerplate code based on architectural diagrams, provide inline code completions with explanations derived from the BMAD context, and allow for easy execution of test suites defined in the workflow. For **testing and review**, the extension would orchestrate the interaction with the BMAD-QA agent, displaying test results, pull request summaries, and vulnerability scans directly in the editor, eliminating the need to switch contexts to external CI/CD dashboards or security scanners. This tight integration ensures that every action taken by a human developer or an AI agent is recorded and reflected in the central BMAD state, maintaining a coherent and auditable history of the project's evolution.

The resulting framework would occupy a unique and valuable position in the market. It would be more structured and reliable than simple prompt-based tools or lightweight frameworks like Spec Kit, making it suitable for professional, team-based development [31]. At the same time, it would be more accessible and easier to adopt than pure BMAD, thanks to the intuitive UI and automation provided by the EMAD-style extension. This positions it as the ideal solution for development teams looking to move beyond "vibe coding" and establish a repeatable, high-quality AI-assisted development practice without sacrificing developer productivity or morale [33]. The framework's ability to combine BMAD's rigorous, spec-driven planning with EMAD's ergonomic, tool-enhanced execution would create a powerful and compelling product. This synthesis is not merely incremental; it represents a step-change in how teams can effectively harness the power of AI, providing the control and predictability of a formal methodology wrapped in the convenience and efficiency of a modern developer tool.

Implementation Considerations and Actionable Recommendations

The successful creation of a new project by integrating the EMAD framework with the BMAD methodology requires careful consideration of several strategic and tactical factors. The analysis confirms that such an integration is not only feasible but also strategically sound, addressing a clear gap in the market for a tool that balances process rigor with developer productivity. To translate this potential into a viable product, the creator of 'Teams for new projects' must proceed with a clear roadmap that addresses critical ambiguities, embraces a hybrid architectural approach, and targets a well-defined audience. The following recommendations outline a path forward based on the comprehensive analysis of the available information.

First and foremost, the project must begin by resolving critical ambiguities present in the initial research goal. The most significant uncertainty is the definition of the "be mad method." The provided sources contain extensive information on the "BMAD Method" [13] [14], but none on a "Be MAD Method." It is imperative to clarify whether this is a typographical error, a reference to a separate, unmentioned philosophy, or a conceptual placeholder. Proceeding without a clear understanding of this core component would jeopardize the project's scope and direction. Similarly, a definitive, hands-on evaluation of the `huggingfacer04/EMAD` framework is non-negotiable. The description "Ecosystem for BMad Method AI Development" is suggestive but lacks concrete details on

its actual features, capabilities, and limitations [1]. A thorough exploration of its source code, documentation, and any available tutorials is required to determine its true nature and identify the specific mechanisms through which it interacts with or complements BMAD. Only with a firm grasp of both constituent parts can a coherent integration strategy be formulated.

Second, the recommended implementation should adopt a hybrid architecture that synthesizes the best attributes of both frameworks. The project's foundation should be the proven, process-oriented engine of the BMAD methodology. This involves implementing BMAD's core principles, including its YAML-driven, stateful workflow management system (`workflow.yaml`) and its role-based, team-simulating agent model [60] [67]. This structural rigor will provide the necessary framework for predictability, quality assurance, and scalability, which are paramount for professional software development teams [33] [102]. However, this underlying structure must be wrapped in a user experience modeled after the EMAD framework. The final product should be designed as a comprehensive VS Code extension, prioritizing developer experience (DX) at every touchpoint [1]. This approach lowers the barrier to entry, reduces friction, and keeps developers within their most productive environment. The extension should serve as an intuitive UI for the BMAD workflow, automating complex command executions, visualizing workflow states, and providing contextual assistance, thereby mitigating BMAD's known complexity and improving discoverability [75].

Third, the project's development should be guided by a clear focus on solving the documented problems of its parent frameworks. The primary innovation opportunity lies in bridging the gap between BMAD's strategic vision and the practical realities of developer workflow. The project must directly address BMAD's complexity by abstracting away the manual management of YAML files and command-line operations. The VS Code extension should provide a visual workflow builder, intelligent prompt generation based on the current state, and automated tool invocation, transforming the potentially daunting process into an interactive and manageable experience. Conversely, if EMAD is a standalone tool, the project should demonstrate a clear and seamless integration path with BMAD, showing how it enhances the execution of the structured methodology. The goal is to create a system where the strengths of each component are amplified, and their weaknesses are neutralized.

Finally, the project should target a specific, well-defined audience to maximize its initial impact. The ideal target market is a software development team or organization that is already exploring AI-assisted development but is struggling with the inconsistencies, inefficiencies, and lack of control inherent in ad-hoc approaches [33] [98]. These are teams

looking to move beyond one-off experiments and institutionalize a reliable, repeatable, and high-quality AI-driven development practice. The proposed framework would appeal to them by offering a structured alternative to "vibe coding" that feels natural to engineers who work within the VS Code ecosystem [24]. Marketing and documentation efforts should emphasize tangible benefits such as improved code quality, accelerated development cycles, and greater team alignment—all powered by a system that respects the developer's existing tools and workflows. By following this strategic path, the creator of 'Teams for new projects' can develop a product that is not merely an incremental update but a genuinely novel and valuable contribution to the field of AI-driven software engineering.

Reference

1. huggingfacer04 https://github.com/huggingfacer04
2. emadbr (Emad Bahrami) https://huggingface.co/emadbr/models
3. Tools for Building LLM-powered Apps | Emad Ibrahim https://www.linkedin.com/posts/eibrahim_viewing-llamaindex-vs-langchain-tools-for-activity-7269366015296167936-CM2b
4. Llamaindex vs Langchain: What's the difference? https://www.ibm.com/think/topics/llamaindex-vs-langchain
5. LangChain vs LlamaIndex https://www.reddit.com/r/LangChain/comments/1bbog83/langchain_vs_llamaindex/
6. LlamaIndex vs LangChain: Which Framework Is Best for ... https://www.zenml.io/blog/llamaindex-vs-langchain
7. LangChain vs LlamaIndex vs AutoGen vs CrewAI https://www.techaheadcorp.com/blog/top-agent-frameworks/
8. LangChain vs LlamaIndex: A Detailed Comparison https://www.datacamp.com/blog/langchain-vs-llamaindex
9. Differences between Langchain & LlamaIndex [closed] https://stackoverflow.com/questions/76990736/differences-between-langchain-llamaindex
10. LangChain vs. LlamaIndex, A Comprehensive Comparison ... https://medium.com/@tam.tamanna18/langchain-vs-llamaindex-a-comprehensive-comparison-for-retrieval-augmented-generation-rag-0adc119363fe

11. LangChain vs LlamaIndex 2025: Complete RAG ... https://latenode.com/blog/platform-comparisons-alternatives/automation-platform-comparisons/langchain-vs-llamaindex-2025-complete-rag-framework-comparison
12. LangChain vs. LlamaIndex. Main differences https://addepto.com/blog/langchain-vs-llamaindex-main-differences/
13. bmad-code-org/BMAD-METHOD https://github.com/bmad-code-org/BMAD-METHOD
14. The BMAD Method: A Framework for Spec Oriented AI- ... https://recruit.group.gmo/engineer/jisedai/blog/the-bmad-method-a-framework-for-spec-oriented-ai-driven-development/
15. Mastering the BMAD Method: A Revolutionary Approach to ... https://medium.com/@courtlinholt/mastering-the-bmad-method-a-revolutionary-approach-to-agile-ai-driven-development-for-modern-e7be588b8d94
16. Anyone here seriously using the BMAD Method for vibe ... https://www.reddit.com/r/vibecoding/comments/1m3b02m/anyone_here_seriously_using_the_bmad_method_for/
17. The Official BMad-Method Masterclass (The Complete IDE ... https://www.reddit.com/r/BMAD_Method/comments/1mge1hz/the_official_bmadmethod_masterclass_the_complete/
18. Bmad Home Page https://www.classe.cornell.edu/bmad/
19. airclear/bmad-method https://www.npmjs.com/package/@airclear/bmad-method
20. BMAD vs. Spek Kit vs. Open Spec: Which AI Coding ... https://www.reddit.com/r/BMAD_Method/comments/1obaopd/bmad_vs_spek_kit_vs_open_spec_which_ai_coding/
21. The Official BMad-Method Masterclass (The Complete IDE ... https://www.youtube.com/watch?v=LorEJPrALcg
22. BMAD + Cursor: The First AI Framework That Actually Makes ... https://www.kevintholland.com/bmad-cursor-the-first-ai-framework-that-actually-makes-sense-for-pms/
23. 24601/BMAD-AT-CLAUDE: Breakthrough Method for Agile ... https://github.com/24601/BMAD-AT-CLAUDE
24. GitHub's BMAD Method Is Ending Vibe Coding https://www.youtube.com/watch?v=t4MBKixQ3fc
25. Ricoledan/bmad-architecture-agent https://github.com/Ricoledan/bmad-architecture-agent
26. bmad-method · GitHub Topics https://github.com/topics/bmad-method

27. A Comparative Analysis of AI Agentic Frameworks: BMAD ... https://medium.com/@mariussabaliauskas/a-comparative-analysis-of-ai-agentic-frameworks-bmad-method-vs-github-spec-kit-edd8a9c65c5e
28. BMAD-METHOD/src/modules/bmb/docs/index.md at main https://github.com/bmad-code-org/BMAD-METHOD/blob/main/src/modules/bmb/docs/index.md
29. BMAD AI/ML Engineering Expansion Pack https://github.com/aisingapore/bmad-aisg-aiml
30. I made a simple comparative analysis ... https://github.com/bmad-code-org/BMAD-METHOD/issues/1219
31. Beyond the Vibe: Why AI Coding Workflows Need a ... https://dzone.com/articles/beyond-vibe-ai-coding-frameworks
32. Agent OS vs BMAD vs Spec Kit – does the framework still ... https://www.reddit.com/r/BMAD_Method/comments/1pcqarr/agent_os_vs_bmad_vs_spec_kit_does_the_framework/
33. Applied BMAD - Reclaiming Control in AI Development https://bennycheung.github.io/bmad-reclaiming-control-in-ai-dev
34. My Recent Discovery of the BMad-Method: Orchestrating AI ... https://www.linkedin.com/pulse/my-recent-discovery-bmad-method-orchestrating-ai-agents-loukine-jkync
35. BMAD vs spec-kit vs OpenSpec vs PromptX https://redreamality.com/blog/-sddbmad-vs-spec-kit-vs-openspec-vs-promptx/
36. BMAD-METHOD | Adam Blackington https://www.adamblackington.com/technical-skills/ai-tools/bmad-method
37. AgentOS VS BMAD VS Spec Kit: Which AI Development ... https://www.youtube.com/watch?v=TxRpWktXBBk
38. Top AI Agent Frameworks of 2025: LangChain, CrewAI & ... https://medium.com/@lambert.watts.809/top-10-best-ai-frameworks-for-building-ai-agents-in-2025-137fafb37a46
39. The Best AI Agent Frameworks For Developers https://www.vellum.ai/blog/top-ai-agent-frameworks-for-developers
40. AI Agent Frameworks: A Detailed Comparison https://www.turing.com/resources/ai-agent-frameworks
41. LangChain vs AutoGen vs CrewAI for Multi-Agent Systems https://www.linkedin.com/pulse/battle-frameworks-langchain-vs-autogen-crewai-multi-agent-7pmjc
42. Best AI Agent Frameworks 2025: LangGraph, CrewAI ... https://www.getmaxim.ai/articles/top-5-ai-agent-frameworks-in-2025-a-practical-guide-for-ai-builders/

43. Top AI Agent Frameworks in 2025 https://www.codecademy.com/article/top-ai-agent-frameworks-in-2025
44. Autogen vs LangChain vs CrewAI https://www.instinctools.com/blog/autogen-vs-langchain-vs-crewai/
45. Langgraph vs CrewAI vs AutoGen vs PydanticAI vs Agno ... https://www.reddit.com/r/LangChain/comments/1jpk1vn/langgraph_vs_crewai_vs_autogen_vs_pydanticai_vs/
46. Top 5 AI Agent Frameworks In 2025 https://www.intuz.com/blog/best-ai-agent-frameworks
47. The Complete Guide to Choosing an AI Agent Framework ... https://www.langflow.org/blog/the-complete-guide-to-choosing-an-ai-agent-framework-in-2025
48. Intelligent Internet Whitepaper - Emad Mostaque https://ii.inc/web/whitepaper
49. JOINT STAFF (EMACON) - EMAD https://emad.defensa.gob.es/en/unidades/emacon/
50. EMAD R Edition 3.0 (05 October 2021) - Approved https://www.fma.no/maanor/downloads/regulations/eda-mawa/EMAD%20R%20(ed%203.1%20%202023-04-21).pdf
51. Prime Intellect - Commoditizing Compute & Intelligence https://www.primeintellect.ai/
52. Software Documentation: Automation and Challenges ... https://www.inf.usi.ch/lanza/Downloads/PhD/Agha2020c.pdf
53. razavioo/emad.dev https://github.com/razavioo/emad.dev
54. Emad Bawah's Post https://www.linkedin.com/posts/emad-bawah-a6b59b1bb_strandsagents-agenticai-aws-activity-7377760751014342657-OwMc
55. Electronic Identity Credential Trust Elevation Framework ... https://docs.oasis-open.org/trust-el/trust-el-framework/v1.0/trust-el-framework-v1.0.html
56. Emad R Edition 3 1 (21 Apr 23 - Approved) https://www.scribd.com/document/644945833/emad-r-edition-3-1-21-apr-23-approved
57. Intelligent Internet - Organizations - IQ.wiki https://iq.wiki/wiki/intelligent-internet
58. Add Scripts Directory for Workflow Automation and Routine ... https://github.com/bmad-code-org/BMAD-METHOD/issues/915
59. Workflow Orchestration: Building Complex AI Pipelines https://medium.com/@omark.k.aly/workflow-orchestration-building-complex-ai-pipelines-c8504ab8306f
60. How to use BMAD v6 for stateful workflows https://www.linkedin.com/posts/martinlionel_who-else-is-usingbmad-v6-i-love-how-activity-7397141343296372736-tf_Y

61. Bmad Orchestrator | PDF | Agile Software Development https://www.scribd.com/document/913367689/Bmad-Orchestrator
62. Building an Agentic Workflow: Orchestrating a Multi-Step ... https://orkes.io/blog/building-agentic-interview-app-with-conductor/
63. BMAD V6 asks to run /bmad:bmm:workflows:create- ... https://github.com/bmad-code-org/BMAD-METHOD/issues/872
64. The Complete Guide to Workflow Orchestration https://kairntech.com/blog/articles/the-complete-guide-to-workflow-orchestration/
65. AI Agile Team Builds a FULL App Step by Step Tutorial ... https://www.youtube.com/watch?v=YLGrENURe98
66. What is Workflow Orchestration? Best Practices + Examples https://www.superblocks.com/blog/workflow-orchestration
67. bmadcode/BMAD-METHOD-v5 https://github.com/bmadcode/BMAD-METHOD-v5
68. bmad-code-org/BMAD-METHOD - Workflow runs https://github.com/bmad-code-org/BMAD-METHOD/actions
69. Suggestion: Simplify the workflow · Issue #555 · bmad- ... https://github.com/bmad-code-org/BMAD-METHOD/issues/555
70. microsoft/task-sequencing-robotics-simulator - Workflow runs https://github.com/microsoft/task-sequencing-robotics-simulator/actions
71. RROS-Lab/IRL-Task-Sequencing https://github.com/RROS-Lab/IRL-Task-Sequencing
72. Failed to parse YAML frontmatter in bmad-workflow-bmm ... https://github.com/bmad-code-org/BMAD-METHOD/issues/1048
73. microsoft/task-sequencing-robotics-simulator https://github.com/microsoft/task-sequencing-robotics-simulator
74. BMAD-METHOD-CH/docs/user-guide.md at main https://github.com/ThomasLXZ/BMAD-METHOD-CH/blob/main/docs/user-guide.md
75. Add VS Code workflow prompt recommendations (chat. ... https://github.com/bmad-code-org/BMAD-METHOD/issues/1206
76. [FEATURE] Asana MCP: Add Section Assignment and ... https://github.com/anthropics/claude-code/issues/10822
77. Activity · microsoft/task-sequencing-robotics-simulator https://github.com/microsoft/task-sequencing-robotics-simulator/activity?sort=ASC
78. Comparing LangGraph, DSPy, CrewAI, Agno, and More https://langwatch.ai/blog/best-ai-agent-frameworks-in-2025-comparing-langgraph-dspy-crewai-agno-and-more

79. LangGraph vs. CrewAI: Which Framework Should You ... https://medium.com/@shashank_shekhar_pandey/langgraph-vs-crewai-which-framework-should-you-choose-for-your-next-ai-agent-project-aa55dba5bbbf
80. What's the best framework for production-grade AI agents ... https://www.reddit.com/r/AI_Agents/comments/1kjowzp/whats_the_best_framework_for_productiongrade_ai/
81. Best Multi-Agent Framework in 2025? Full Head-to- ... https://www.youtube.com/watch?v=KDiO7Zg2N0E
82. LangGraph vs CrewAI: Comparison Guide for Production ... https://xcelore.com/blog/langgraph-vs-crewai/
83. Top 7 AI Agent Frameworks in 2025 — Ultimate Guide https://www.ampcome.com/post/top-7-ai-agent-frameworks-in-2025
84. Comparing AI Agent Frameworks: Which One Should I ... https://community.latenode.com/t/comparing-ai-agent-frameworks-which-one-should-i-choose-for-my-project/31007
85. Emad-COMBINE-lab https://github.com/Emad-COMBINE-lab
86. emad-elsaid/emad-elsaid.github.io: My personal Blog https://github.com/emad-elsaid/emad-elsaid.github.io
87. Seyyed Emad Razavi | Senior Mobile Developer https://emad.dev/
88. Emad-Gamal https://github.com/Emad-Gamal
89. Context-Aware Software Documentation https://ieeexplore.ieee.org/servlet/Login?logout=/document/8530093/
90. Consolidated Planning/CHAS Data https://www.huduser.gov/portal/datasets/cp.html
91. Emad Mostaque, Founder/CEO Stability AI (TRANSCRIPT) https://sarahguo.com/blog/emadmostaque
92. Emad Shawky's Post https://www.linkedin.com/posts/emad-shawky-8904421a_best-book-ive-read-in-the-last-couple-of-activity-7275052863632928768-WbVY
93. An Empirical Study of Agent Developer Practices in AI ... https://arxiv.org/html/2512.01939v1
94. An empirical evaluation of technology acceptance model ... https://www.sciencedirect.com/science/article/pii/S2405844023055573
95. From Hype to Hands-On: How to Build and Launch AI Agents ... https://www.youtube.com/watch?v=CkgYrkw0iwo
96. Exploring the Readiness of Organisations to Adopt Artificial ... https://www.researchgate.net/publication/383004893_Exploring_the_Readiness_of_Organisations_to_Adopt_Artificial_Intelligence

97. Facilitating adoption of responsible innovation in business ... https://www.tandfonline.com/doi/full/10.1080/23299460.2023.2211810
98. Emad might've just figured out how a new economic ... https://www.instagram.com/reel/DMxev0ShoVO/
99. 2: The current state of the generative artificial intelligence ... https://www.elgaronline.com/monochap/book/9781035346745/chapter2.xml
100. Gen AI is now a daily tool for enterprises https://www.facebook.com/groups/aifire.co/posts/1849884002283567/
101. A Look into the BMAD Method. - SoloDev.app https://solodev.app/a-look-into-the-bmad-method
102. How BMAD Method Makes AI Development Deterministic https://www.linkedin.com/pulse/how-bmad-method-makes-ai-development-deterministic-lessons-hofer-bfbce
103. Exploring the BMAD METHOD for Systematic AI Development https://python.plainenglish.io/from-specifications-to-context-engineering-exploring-the-bmad-method-for-systematic-ai-development-4346d4da2b18
104. Diving Into Spec-Driven Development With GitHub Spec Kit https://developer.microsoft.com/blog/spec-driven-development-spec-kit
105. BMAD: The AI Development Secret No One's Talking About ... https://medium.com/@visrow/bmad-the-ai-development-secret-no-ones-talking-about-actually-everyone-is-9c1cabce8a10
106. BMAD: 6-Step AI Dev Workflow for Structured Code Success https://www.theaistack.dev/p/bmad
107. Emad-COMBINE-lab/MARSY https://github.com/Emad-COMBINE-lab/MARSY
108. imaginAIry/docs/emad-qa-2020-10-10.md at master https://github.com/brycedrennan/imaginAIry/blob/master/docs/emad-qa-2020-10-10.md
109. Mohammed Emad | AI & Machine Learning Engineer https://github.com/mohammed-Emad
110. youssefemadeldn/Manfaz-Client https://github.com/youssefemadeldn/Manfaz-Client
111. Gumiho: A Hybrid Architecture to Prioritize Early Tokens in ... https://github.com/AMD-AIG-AIMA/Gumiho
112. OpenMOSS/Thus-Spake-Long-Context-LLM https://github.com/OpenMOSS/Thus-Spake-Long-Context-LLM
113. README.md - swyxio/ai-notes https://github.com/swyxio/ai-notes/blob/main/README.md
114. Awesome-Efficient-LLM/pruning.md at main https://github.com/horseee/Awesome-Efficient-LLM/blob/main/pruning.md

115. EmadBeltaje/flutter_getx_template: Create flutter project ... https://github.com/EmadBeltaje/flutter_getx_template
116. Kubeflow vs. MLflow: An In-Depth Comparison for MLOps ... https://skphd.medium.com/kubeflow-vs-mlflow-an-in-depth-comparison-for-mlops-pipelines-e2f0c0496a36
117. A Brief Comparison of Kubeflow vs. MLflow https://jfrog.com/blog/a-brief-comparison-of-kubeflow-vs-mlflow/
118. MLflow vs Kubeflow: Comparison for AI/ML Practitioners https://www.linkedin.com/pulse/mlflow-vs-kubeflow-in-depth-comparison-aiml-nagesh-deshmukh-1m8sf
119. Kubeflow vs MLflow vs ZenML: Which MLOps Platform Is ... https://www.zenml.io/blog/kubeflow-vs-mlflow
120. A Comprehensive Comparison Between Kubeflow and ... https://valohai.com/blog/kubeflow-vs-mlflow/
121. Kubeflow vs MLFlow https://www.youtube.com/watch?v=Eve6WVXURoQ
122. Choosing the Right MLOps Framework for Scalable AI https://www.techugo.com/blog/kubeflow-vs-mlflow-choosing-the-right-mlops-framework-for-scalable-ai/
123. MLflow vs Kubeflow : r/mlops https://www.reddit.com/r/mlops/comments/1evza42/mlflow_vs_kubeflow/
124. Kubeflow VS MLflow Comparison in MLOps https://divedeep.ai/kubeflow-vs-mlflow-comparison-in-mlops/
125. Compare Kubeflow vs. MLflow | Blog https://superwise.ai/blog/kubeflow-vs-mlflow/
126. The AI Agent Framework Landscape in 2025 https://medium.com/@hieutrantrung.it/the-ai-agent-framework-landscape-in-2025-what-changed-and-what-matters-3cd9b07ef2c3
127. Tested 5 agent frameworks in production - here's when to ... https://www.reddit.com/r/AI_Agents/comments/1oukxzx/tested_5_agent_frameworks_in_production_heres/
128. AutoGen vs CrewAI vs LangGraph: AI Framework https://jetthoughts.com/blog/autogen-crewai-langgraph-ai-agent-frameworks-2025/
129. Best Agentic AI Frameworks 2025: LangGraph, AutoGen ... https://www.linkedin.com/pulse/best-agentic-ai-frameworks-2025-langgraph-autogen-crewai-ambatwar-kiltf
130. LangGraph vs AutoGen vs CrewAI: Complete AI Agent ... https://latenode.com/blog/platform-comparisons-alternatives/automation-platform-comparisons/langgraph-vs-autogen-vs-crewai-complete-ai-agent-framework-comparison-architecture-analysis-2025
131. NIST.SP.800-63-3.pdf https://canada-ca.github.io/PCTF-CCP/docs/RelatedPolicies/NIST.SP.800-63-3.pdf

132. What do Programmers Discuss about Deep Learning ... https://xin-xia.github.io/publication/emse195.pdf

133. The Impact of Environment Configurations on the Stability ... https://aabdllatif.github.io/assets/pdf/UTF-8rahman_ease2025.pdf

134. COLLECTIVE VARIABLES MODULE Reference manual for ... https://colvars.github.io/master/colvars-refman-namd.pdf

135. nist.sp.800-63b.pdf https://canada-ca.github.io/PCTF-CCP/docs/RelatedPolicies/NIST.SP.800-63b.pdf

136. Availability and Usage of Platform-Specific APIs https://andrehora.github.io/pub/2024-msr-ptaform-specific-apis.pdf

137. DeepResearch Bench: A Comprehensive Benchmark for ... https://deepresearch-bench.github.io/static/papers/deepresearch-bench.pdf

138. Generating Realistic Vulnerabilities via Neural Code Editing https://chapering.github.io/pubs/fse22yu.pdf

139. Emad Saad Alsuwat, Ph.D. Associate Professor https://emadalsuwat.github.io/resources/cv.pdf

140. A Comprehensive Study on Challenges in Deploying Deep ... https://taoxiease.github.io/publications/esecfse20-deploydl.pdf

141. emadeldeen24/CoTMix: [TAI 2023] Contrastive Domain ... https://github.com/emadeldeen24/CoTMix

142. GitHub marketplace for automation and innovation in ... https://www.sciencedirect.com/science/article/pii/S0950584924001277

143. Feature: Generate AMD output · Issue #819 · evanw/esbuild https://github.com/evanw/esbuild/issues/819

144. GitHub Marketplace for Automation and Innovation in ... https://arxiv.org/pdf/2407.05519

145. Studying the Impact of TensorFlow and PyTorch Bindings ... https://dl.acm.org/doi/full/10.1145/3678168

146. awesome-mlops/papers.md at master https://github.com/visenger/awesome-mlops/blob/master/papers.md

147. SemanticOn: Specifying Content-Based Semantic ... https://chensivan.github.io/papers/uist22_semanticon.pdf

148. There's a reason Stripe doesn't charge per user and Slack ... https://www.linkedin.com/posts/roblitterst_theres-a-reason-stripe-doesnt-charge-per-activity-7274815640857710592-EX2W

149. Studying the Security and Maintainability of MCP Servers https://www.researchgate.net/publication/

392736704_Model_Context_Protocol_MCP_at_First_Glance_Studying_the_Security_and_Maintainability_of_MCP_Servers

150. A Comprehensive Survey on Segment Anything Model for ... https://github.com/liliu-avril/Awesome-Segment-Anything
151. hemingkx/SpeculativeDecodingPapers: Must-read ... https://github.com/hemingkx/SpeculativeDecodingPapers
152. Xnhyacinth/Awesome-LLM-Long-Context-Modeling https://github.com/Xnhyacinth/Awesome-LLM-Long-Context-Modeling
153. lobehub/lobe-chat-agents: / Agent Index https://github.com/lobehub/lobe-chat-agents
154. swyxio/ai-notes https://github.com/swyxio/ai-notes
155. APRIL: Active Partial Rollouts in Reinforcement Learning to ... https://github.com/RLsys-Foundation/APRIL
156. Top 8 architecture for building AI models Follow Data ... https://www.facebook.com/groups/2059467967664033/posts/4177964332481042/
157. Introducing BuildPiper MCP Server: AI for DevSecOps https://www.linkedin.com/posts/shankar-prasad-jha_buildpiper-mcp-productivity-activity-7353322460219367425-Zswv
158. part i: concepts and design thinking innovation addressing ... https://library.oapen.org/bitstream/id/5cf3022e-b665-449c-b9f5-75dc22bed290/9781638282297.pdf
159. Here's how you can kickstart your AI learning journey as a ... https://www.linkedin.com/advice/1/heres-how-you-can-kickstart-your-ai-learning-ncp0c
160. My product team at Pawa IT Solutions is hiring a full stack ... https://www.facebook.com/groups/icthubkenya/posts/3784856328324359/
161. We're thrilled to announce that John Snow https://www.facebook.com/groups/1488324504555672/posts/8791241650930551/
162. Building Custom AI Agents with BMB and Google AntiGravity https://medium.com/@visrow/bmad-method-building-custom-ai-agents-with-bmb-and-google-antigravity-54ac96024e94
163. Agentic Software Engineering: Foundational Pillars and a ... https://arxiv.org/html/2509.06216v1
164. emad-elsaid https://github.com/emad-elsaid
165. Emadeldeen Eldele emadeldeen24 https://github.com/emadeldeen24
166. emad-zaamout/laravel-websockets-chat-application-example https://github.com/emad-zaamout/laravel-websockets-chat-application-example

167. opendatacube/odc-stats: Statistician is a framework ... https://github.com/opendatacube/odc-stats
168. facebookresearch/matrix: Matrix (Multi-Agent daTa ... https://github.com/facebookresearch/matrix
169. mayuelala/Awesome-Controllable-Video-Generation ... https://github.com/mayuelala/Awesome-Controllable-Video-Generation
170. ryanbgriffiths/ICRA2024PaperList: ICRA2024 Paper List https://github.com/ryanbgriffiths/ICRA2024PaperList
171. ROCm/MAD: MAD (Model Automation and Dashboarding) https://github.com/ROCm/MAD
172. An Exploratory Study on Machine Learning Model ... https://dl.acm.org/doi/full/10.1145/3688841
173. The Impact of Code Ownership of DevOps Artefacts on the ... https://nimmiw.com/papers/MSR_DevOps_Code_Ownership__CR.pdf
174. Perception of Software Bots on Pull Requests on Social ... https://mairieli.com/files/papers/WesselBots.pdf
175. The UK in 2024 Phase One AI and Open Innovation https://openuk.uk/wp-content/uploads/2024/02/State-of-Open-The-UK-in-2024-Phase-One.pdf
176. Diganta Misra https://digantamisra98.github.io/
177. Emad Adly EmadAdly https://github.com/emadadly
178. SamuelSchmidgall/AgentLaboratory https://github.com/SamuelSchmidgall/AgentLaboratory
179. khaledsabry97/Argus: [IEEE Research paper ... https://github.com/khaledsabry97/Argus
180. Xinglab/CLAM: CLIP-seq Analysis of Multi-mapped reads https://github.com/Xinglab/CLAM
181. GitHub - emadeldeen24/TS-TCC: [IJCAI-21] "Time-Series ... https://github.com/emadeldeen24/TS-TCC
182. Open Source Data Engineering Landscape 2024 https://www.pracdata.io/p/open-source-data-engineering-landscape
183. [Biomedical Signal Processing and Control] ECGTransForm https://github.com/emadeldeen24/ECGTransForm
184. Vulnerabilities from GitHub Repositories Hosted by Tech Orgs https://dl.acm.org/doi/fullHtml/10.1145/3546096.3546114
185. Predicting GitHub Contributors with Machine Learning https://www.sciencedirect.com/science/article/pii/S0950584921000902

186. [IEEE TETCI] "ADAST: Attentive Cross-domain EEG-based ... https://github.com/emadeldeen24/ADAST

187. Dependabot's Impact on Vulnerability Mitigation Study https://link.springer.com/article/10.1007/s10664-025-10638-w

188. Generative AI for Software Architecture. Applications, ... https://arxiv.org/html/2503.13310v2

189. Emad 's response about Stable Diffusion 3 in ... https://www.facebook.com/groups/stablediffusionkorea/posts/1556833395159502/

190. A Hardware Design Framework for Computer Vision ... https://dl.acm.org/doi/full/10.1145/3635157

191. Akhil Sharma's Post https://www.linkedin.com/posts/akhilsails_%F0%9D%90%87%F0%9D%90%9E%F0%9D%90%AB%F0%9D%90%9E%F0%9D%90%AC-%F0%9D%90%B0%F0%9D%90%A1%F0%9D%90%9A%F0%9D%90%AD-%F0%9D%90%A7%F0%9D%90%A8%F0%9D%90%9B%F0%9D%90%A8%F0%9D%90%9D%F0%9D%90%B2-%F0%9D%90%AD%F0%9D%90%9E-activity-7388809258005483520-tJww

192. CVPR 2025 Awards https://cvpr.thecvf.com/virtual/2025/awards_detail

193. Improving software development life cycle using data- ... https://www.eurecom.eu/publication/7682/download/sec-publi-7682.pdf

194. Screenshot of Example Workflow and Adaptation Rules in ... https://www.researchgate.net/figure/Screenshot-of-Example-Workflow-and-Adaptation-Rules-in-jBoss-Drools_fig1_221386526

195. Software Engineering by and for Humans in an AI Era - Monash https://researchmgt.monash.edu/ws/portalfiles/portal/731898981/709055507-oa.pdf

196. Model Context Protocol (MCP): Landscape, Security ... https://xinyi-hou.github.io/files/hou2025mcp_1.pdf

197. Ontology Handbook-Cover https://www.cto.mil/wp-content/uploads/2025/06/SERC_Handbook-on-Digital-Engineering-with-Ontologies_2.0.pdf

198. (PDF) Agentic Software Engineering: Foundational Pillars ... https://www.researchgate.net/publication/395355277_Agentic_Software_Engineering_Foundational_Pillars_and_a_Research_Roadmap

199. World Bank Document https://documents.worldbank.org/curated/en/865481468004211871/pdf/multi0page.pdf

200. AMD https://github.com/amd

201. Open edX https://github.com/openedx

202. Discussion of 10 Standout GitHub Profile READMEs https://dev.to/github/10-standout-github-profile-readmes-h2o/comments

203. DSDGen: Extracting Documentation to Comprehend Fork ... https://jacobkrueger.github.io/assets/papers/Krueger2023DSDGen.pdf

204. GitHub - xm1k3/cent: Community nuclei templates https://github.com/xm1k3/cent

205. DSDGen: Extracting Documentation to Comprehend Fork ... https://dl.acm.org/doi/pdf/10.1145/3579028.3609015

206. How Are GitHub Communication Channels Presented? Analysis https://arxiv.org/pdf/2205.01440

207. Build B2B integrations with AI and Ampersand https://www.linkedin.com/posts/jatin-sandilya-87914893_nowyoucanbuildproduction-gradeb2bintegrations-activity-7321243719318310912-QB9C

208. Mochi Media, Inc. https://github.com/mochi

209. FatimahEmadEldin/Constrained-Track-Document-Bassline ... https://huggingface.co/FatimahEmadEldin/Constrained-Track-Document-Bassline-Readability-Arabertv2-d3tok-reg

210. Automating Code Generation from Scientific Papers in ... https://arxiv.org/html/2504.17192v2

211. An Empirical Study of Developer-Provided Context for AI ... https://arxiv.org/pdf/2512.18925

212. AI-coupled HPC Workflow Applications, Middleware and ... https://arxiv.org/html/2406.14315v2

213. Automating Code Generation from Scientific Papers in ... https://arxiv.org/pdf/2504.17192

214. LLM-based Agentic Reasoning Frameworks: A Survey from ... https://arxiv.org/html/2508.17692v1

215. Retrieval-Augmented Code Generation: A Survey with ... https://arxiv.org/html/2510.04905v1

216. A Comprehensive Survey of Deep Research: Systems ... https://arxiv.org/html/2506.12594v1

217. A Compound AI Architecture for Scientific Discovery https://arxiv.org/html/2511.18298v1

218. Generative AI for Self-Adaptive Systems https://arxiv.org/pdf/2512.04680

219. Model Context Protocol (MCP): Landscape, Security ... https://arxiv.org/html/2503.23278

220. 3DGS and Beyond Docs https://github.com/yangjiheng/3DGS_and_Beyond_Docs

221. Summarization-Papers/README.md at main https://github.com/xcfcode/Summarization-Papers/blob/main/README.md
222. zhusiling/super-resolution-with-GAN https://github.com/zhusiling/super-resolution-with-GAN
223. ryanbgriffiths/ICRA2023PaperList: ICRA2023 Paper List https://github.com/ryanbgriffiths/ICRA2023PaperList
224. xcfcode/Summarization-Papers https://github.com/xcfcode/Summarization-Papers
225. theguly/stars https://github.com/theguly/stars
226. fabiochiusano/Awesome-AI-News https://github.com/fabiochiusano/Awesome-AI-News
227. Emad Ghorbaninia https://emadgnia.github.io/
228. Agent Laboratory https://agentlaboratory.github.io/
229. Exploring Differences and Commonalities between Feature ... https://cmustrudel.github.io/papers/meinicke20flags.pdf
230. Reactive PostgreSQL Client https://vertx-web-site.github.io/docs/vertx-pg-client/java/
231. Is Using Deep Learning Frameworks Free?Characterizing ... https://xin-xia.github.io/publication/icse201.pdf
232. Research Projects https://neurodatascience.github.io/research.html
233. autonomous-workflow-agent-architecture.md https://github.com/nibzard/awesome-agentic-patterns/blob/main/patterns/autonomous-workflow-agent-architecture.md
234. Microsoft Agent Framework: An Open-Source Kit for AI ... https://www.linkedin.com/posts/emadadel_introduction-to-microsoft-agent-framework-activity-7379214487595237376-kZLM
235. Example Workflow Orchestrator implementation https://github.com/workfloworchestrator/example-orchestrator
236. Philippe Bogaerts' Post - microsoft/agent-framework https://www.linkedin.com/posts/philippebogaerts_github-microsoftagent-framework-a-framework-activity-7379602364057485313-RSZK
237. Create an API-driven resource orchestration framework ... https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/create-an-api-driven-resource-orchestration-framework-using-github-actions-and-terragrunt.html
238. AMD-AGI/Primus https://github.com/AMD-AGI/Primus
239. Tip 1: Build your first GitHub Actions workflow https://www.youtube.com/watch?v=47zYGHwXPmE

240. workflow-orchestrator.md - awesome-claude-code-subagents https://github.com/VoltAgent/awesome-claude-code-subagents/blob/main/categories/09-meta-orchestration/workflow-orchestrator.md
241. BMAD-METHOD/src/modules/bmm/docs/index.md at main https://github.com/bmad-code-org/BMAD-METHOD/blob/main/src/modules/bmm/docs/index.md
242. How BMAD v6 Revolutionized AI-Assisted Development https://medium.com/@hieutrantrung.it/from-token-hell-to-90-savings-how-bmad-v6-revolutionized-ai-assisted-development-09c175013085
243. BMAD v6 : The Future of Coding. https://www.youtube.com/watch?v=Zo6_d_xdgRY
244. he BMAD Builder is Here - V6 Alpha: 5 Major Features https://www.reddit.com/r/BMAD_Method/comments/1o01mhn/he_bmad_builder_is_here_v6_alpha_5_major_features/
245. BMAD: AI-Powered Agile Framework Overview - Plaban Nayak https://nayakpplaban.medium.com/bmad-ai-powered-agile-framework-overview-238d4af39aa4
246. BMAD-METHOD - Universal AI Agent Framework Tutorial Guide https://bmadmethodguide.com/
247. BMAD Method - A completely new way to develop Software. https://www.youtube.com/watch?v=JuyIdqa0HJM
248. 50kB https://gitingest.com/bmad-code-org/BMAD-METHOD
249. (1/2) What's the Best AI Framework: LangChain ... - Medium https://medium.com/@cubode/whats-the-best-ai-framework-langchain-or-llamaindex-75da11b917d9
250. Langchain vs LlamaIndex vs CrewAI vs Custom? Which ... https://www.reddit.com/r/LocalLLaMA/comments/1chkl62/langchain_vs_llamaindex_vs_crewai_vs_custom_which/
251. LangChain vs LangGraph vs LlamaIndex: Best LLM ... https://xenoss.io/blog/langchain-langgraph-llamaindex-llm-frameworks
252. How do I integrate LlamaIndex with other libraries like ... https://milvus.io/ai-quick-reference/how-do-i-integrate-llamaindex-with-other-libraries-like-langchain-and-haystack
253. Comparison of LLM Frameworks and Their Capabilities https://www.facebook.com/groups/DeepNetGroup/posts/2299985437060992/
254. Which framework should I choose for RAG development https://community.latenode.com/t/which-framework-should-i-choose-for-rag-development-langchain-vs-llamaindex/39107
255. LlamaIndex - Redefine Document Workflows with AI Agents https://www.llamaindex.ai/

256. Awesome list for LLM quantization https://github.com/pprp/Awesome-LLM-Quantization
257. EMAD - A deployment tool that uses rsync and sshfs https://github.com/ema-digital/emad
258. Future of this repo - I found promising fork ... https://github.com/AUTOMATIC1111/stable-diffusion-webui/discussions/9505
259. Use Multiple Repositories | Adobe Experience Manager https://experienceleague.adobe.com/en/docs/experience-manager-cloud-service/content/implementing/using-cloud-manager/managing-code/working-with-multiple-source-git-repositories
260. aem-project-content-package-structure.md https://github.com/AdobeDocs/experience-manager-cloud-service.en/blob/main/help/implementing/developing/introduction/aem-project-content-package-structure.md
261. Setting up your development environment https://github.com/adaptlearning/adapt_framework/wiki/Setting-up-your-development-environment
262. EASILY Setup Auto-Deployment with Render & Github https://www.youtube.com/watch?v=TE0aQ4csI0o
263. git repository structure with multiple modules https://stackoverflow.com/questions/5795089/git-repository-structure-with-multiple-modules
264. Managing Multi-Repo AEM Projects in AEM as a Cloud ... https://pradeepmoolemane.medium.com/managing-multi-repo-aem-projects-in-aem-as-a-cloud-service-using-git-submodules-df5ec0ef523a
265. Tutorial 1- End To End ML Project With Deployment-Github ... https://www.youtube.com/watch?v=Rv6UFGNmNZg
266. How To Structure Your Git Repository For DevOps ... https://octopus.com/blog/devops-automation-repo-design
267. BMad Method V6 Quick Start Guide https://github.com/bmad-code-org/BMAD-METHOD/blob/main/docs/modules/bmm-bmad-method/quick-start.md
268. BMAD-METHOD/docs/index.md at main https://github.com/bmad-code-org/BMAD-METHOD/blob/main/docs/index.md
269. The BMAD Method: The Ultimate AI Coding System https://www.youtube.com/watch?v=fD8NLPU0WYU&vl=en
270. Spec-Driven Development: The Waterfall Strikes Back https://marmelab.com/blog/2025/11/12/spec-driven-development-waterfall-strikes-back.html
271. Getting started with BMAD-METHOD. Introduction: https://medium.com/@vaishnavikumarpandian/getting-started-with-bmad-method-ccfdb88fb3d6

272. Combining LangChain and LlamaIndex: A Practical Guide ... https://medium.com/@adilmaqsood501/combining-langchain-and-llamaindex-a-practical-guide-with-code-4b988f38217b

273. LlamaIndex | Integration guides https://www.llama.com/docs/integration-guides/llamaindex/

274. use LlamaIndex & LangChain together |tutorial:8 https://www.youtube.com/watch?v=MDxlVnAsvUk

275. LLamaindex and Langchain Integration https://aimarketplace.co/llamaindex-and-langchain-integration

276. Serving A LlamaIndex RAG App as REST APIs https://www.bentoml.com/blog/serving-a-llamaindex-rag-app-as-rest-apis

277. How to Connect LlamaIndex with Private LLM API ... https://www.dataleadsfuture.com/how-to-connect-llamaindex-with-private-llm-api-deployments/

278. Integration with LangChain and LlamaIndex — ACE Agent https://docs.nvidia.com/ace/ace-agent/4.1/user/integrate-langchan-llmaindex.html

279. Intelligent-Internet/CommonGround https://github.com/Intelligent-Internet/CommonGround

280. csyxwei/Awesome-Personalized-Image-Generation https://github.com/csyxwei/Awesome-Personalized-Image-Generation

281. CompVis/stable-diffusion: A latent text-to-image ... https://github.com/CompVis/stable-diffusion

282. Awesome-LLM-Long-Context-Modeling/README.md at main https://github.com/Xnhyacinth/Awesome-LLM-Long-Context-Modeling/blob/main/README.md

283. ICRA2024PaperList/README.md at main https://github.com/ryanbgriffiths/ICRA2024PaperList/blob/main/README.md

284. In-depth tutorials on LLMs, RAGs and real-world AI agent ... https://www.facebook.com/groups/devtitans/posts/1148017390831778/

285. Newest 'agent' Questions https://stackoverflow.com/questions/tagged/agent?tab=Newest

286. A Comprehensive Survey of Deep Research https://arxiv.org/pdf/2506.12594?

287. Topic: fine-tuning https://news.smol.ai/tags/fine-tuning/

288. An Exploration of Using Retrieval-Augmented Generation ... https://trepo.tuni.fi/bitstream/handle/10024/159017/ChenBingxiang.pdf?sequence=3&isAllowed=y

289. How AI turned 500K procurement records into real-time insights https://www.linkedin.com/posts/mohamed-farag-3169b5246_ai-multiagentsystems-llm-activity-7352387098034745344-61jh

290. Model Context Protocol (MCP): Landscape, Security ... https://www.researchgate.net/publication/390354015_Model_Context_Protocol_MCP_Landscape_Security_Threats_and_Future_Research_Directions
291. Annual Meeting of the Association for Computational ... https://aclanthology.org/events/acl-2025/
292. Book overview “This is the most comprehensive textbook ... https://www.facebook.com/groups/899723020138800/posts/9264336357010716/
293. Basic Raw Data Exploration https://www.kaggle.com/code/asaniczka/basic-raw-data-exploration
294. Optimizing Foundation Models: LoRA, MoE, and Beyond https://medium.com/@emad.ghadirian/optimizing-foundation-models-lora-moe-and-beyond-54c0c0964e19
295. An Empirical Study of Context Files for Agentic Coding https://arxiv.org/html/2511.12884v1
296. Lifecycle-Aware code generation: Leveraging Software ... https://arxiv.org/html/2510.24019v1
297. Scaling Agentic Kernel Coding for Heterogeneous AI ... https://arxiv.org/html/2512.23236
298. DocAgent: A Multi-Agent System for Automated Code ... https://arxiv.org/html/2504.08725v1
299. From Prompts to Payments: Unlocking LLMs with Stripe ... https://emaddehnavi.medium.com/from-prompts-to-payments-unlocking-llms-with-stripe-integration-a729c0a7b7bd
300. ScheduleMe: Multi-Agent Calendar Assistant https://arxiv.org/pdf/2509.25693
301. Newest 'langchain' Questions https://stackoverflow.com/questions/tagged/langchain?tab=Newest
302. How to evolve your business model with apps | Emad Medher ... https://www.linkedin.com/posts/emadmedher_the-safest-one-for-evolving-an-business-model-activity-7337397514704203776-Fn8q
303. A Comprehensive Study of Technical Debt Across LLM ... https://dl.acm.org/doi/10.1145/3756681.3756976
304. ScheduleMe: Multi-Agent Calendar Assistant https://arxiv.org/html/2509.25693v2
305. Hey Question. What do you need to build a good AI ... https://www.facebook.com/groups/aibusinesstools/posts/1635429811196277/
306. Trending 'langchain' questions https://stackoverflow.com/questions/tagged/langchain?tab=Trending

307. 34.9 kB https://huggingface.co/TheBloke/WhiteRabbitNeo-33B-v1-GGUF/resolve/main/README.md?download=true

308. MASLab: A Unified and Comprehensive Codebase for LLM ... https://arxiv.org/html/2505.16988v1

309. A Survey on Efficient Architectures for Large Language ... https://arxiv.org/html/2508.09834v1

310. An Empirical Study of Context Files for Agentic Coding https://arxiv.org/pdf/2511.12884

311. Audit-LLM: Multi-Agent Collaboration for Log-based Insider ... https://arxiv.org/html/2408.08902v1

312. Model Context Protocol (MCP): Landscape, Security ... https://arxiv.org/html/2503.23278v3

313. envbench:abenchmark for automated environment setup https://arxiv.org/pdf/2503.14443